

Dual Slope converter:

- # Adv: It requires small amount of circuit components. We can use many levels for quantization.
- # Disadvantage: Relatively slow.

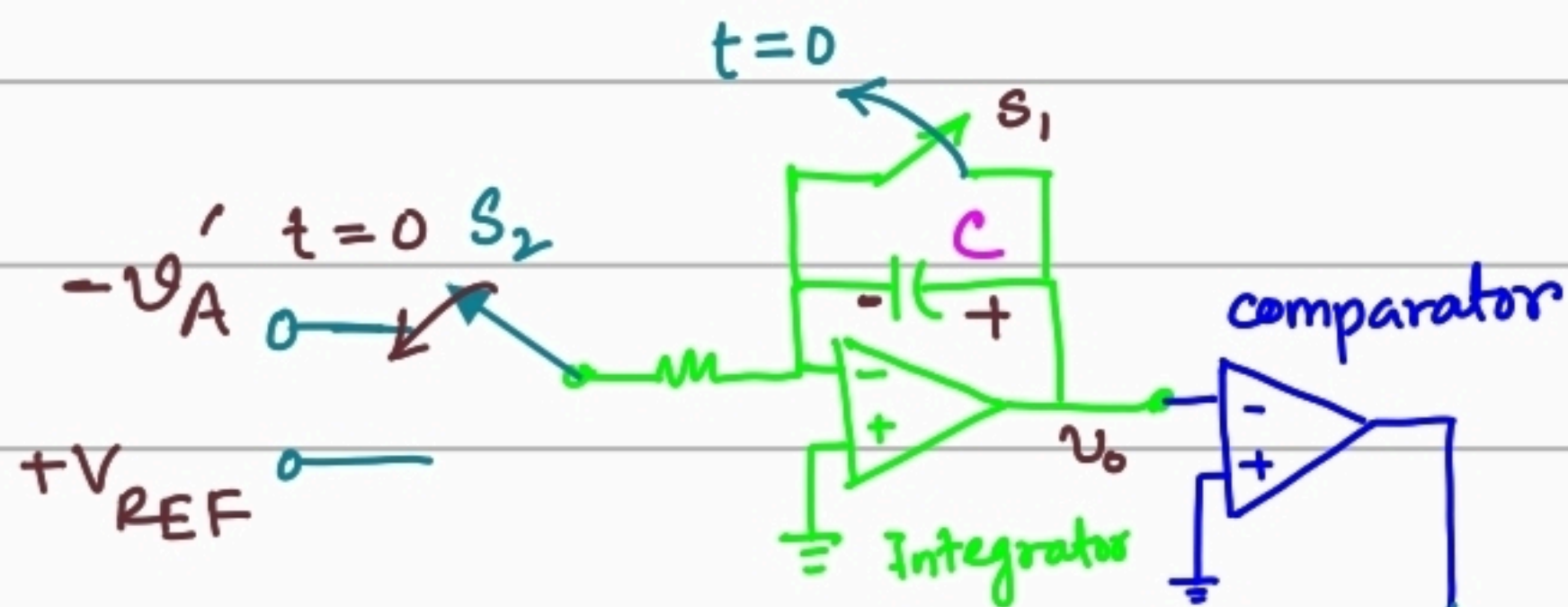
Circuit:

V_A' = Sampled and held signal

V_{REF} = preassigned value.

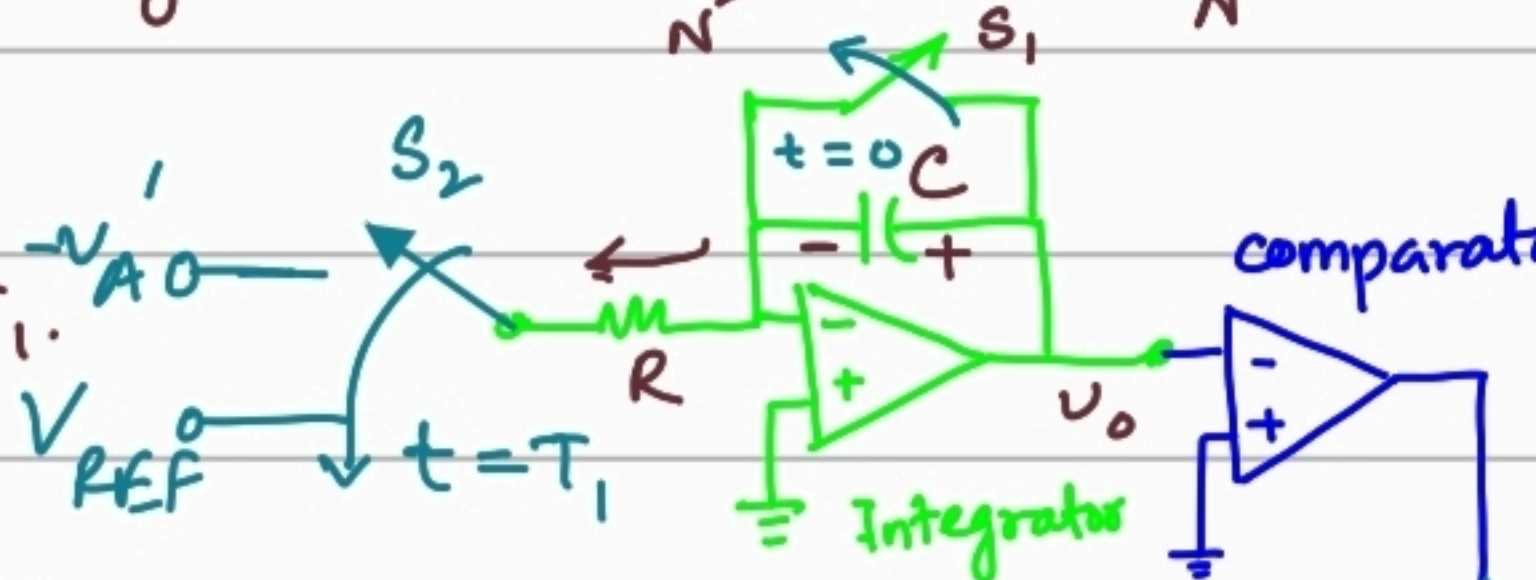
□ ($t < 0$) S_1 is closed and S_2 is open.

Capacitor doesn't have any charge.



□ $t = 0$, S_1 is open. S_2 is now connected to V_A' .
Capacitor is charging in a way that V_o increases.
Counter starts counting from $\underbrace{000\dots 0}_N$ to $\underbrace{111\dots 1}_N$.

□ $t \geq T_1$. Counter finishes counting from 0 to $2^N - 1$.
at $t = T_1$.
 S_2 is now connected to V_{REF} . From then capacitor discharges and V_o starts to decline.



□ $t = T_2$ (not fixed time)

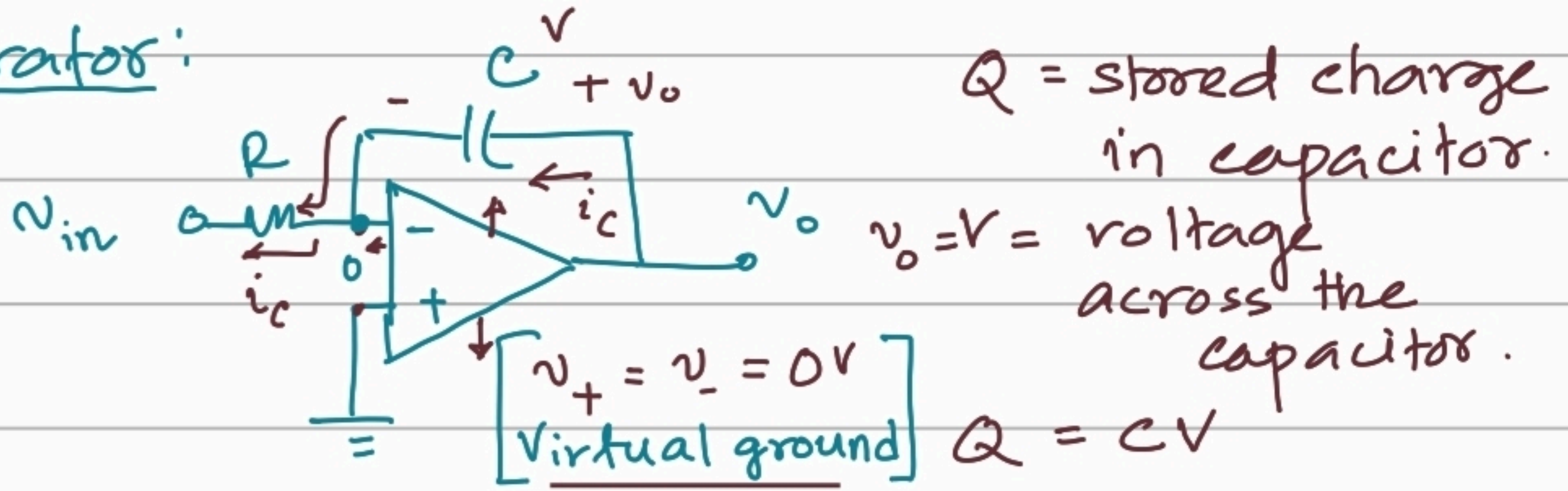
V_o output level decreases below 0V.

Now comparator output changes from low to high.

At that moment control circuit stops the counter.

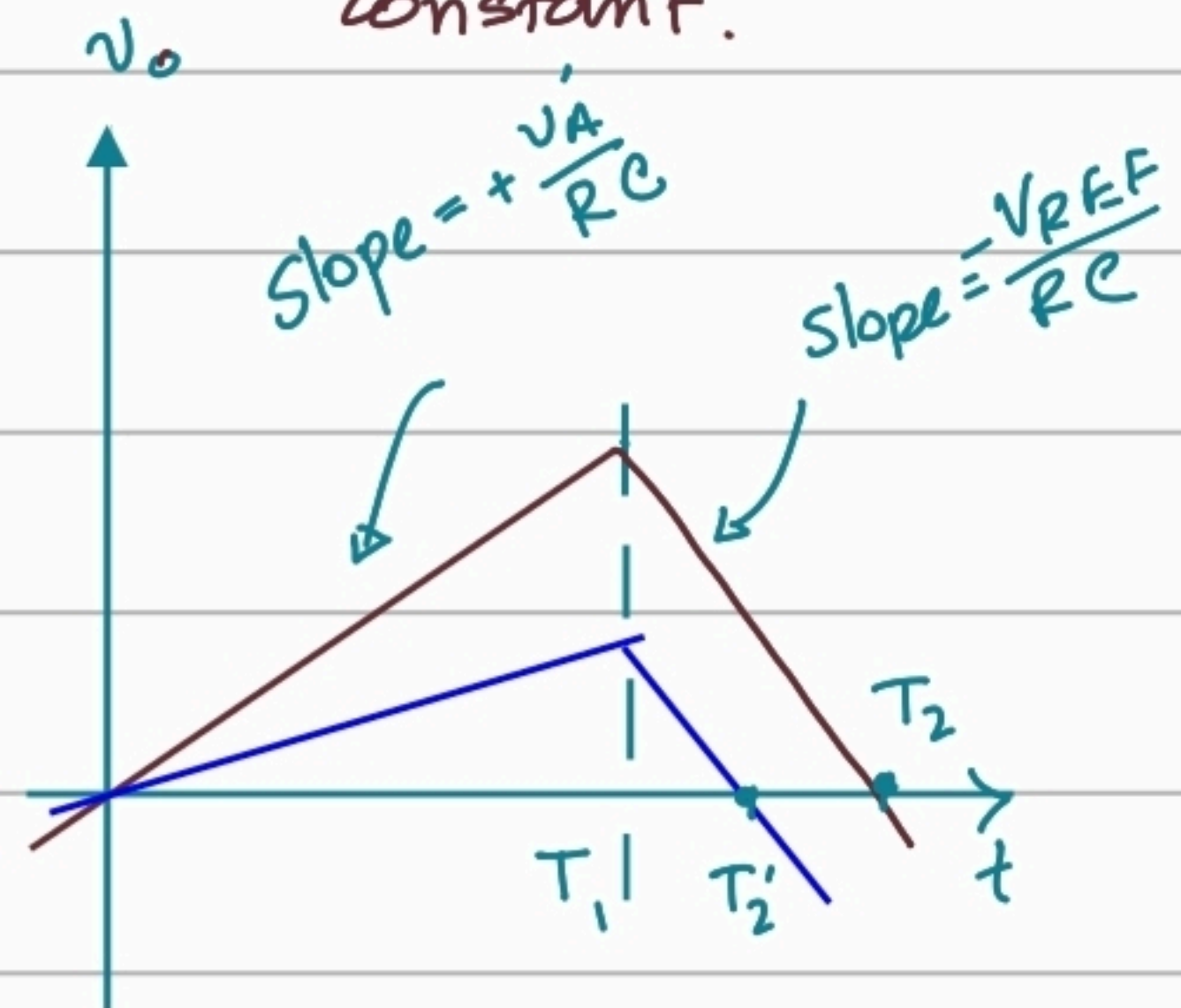
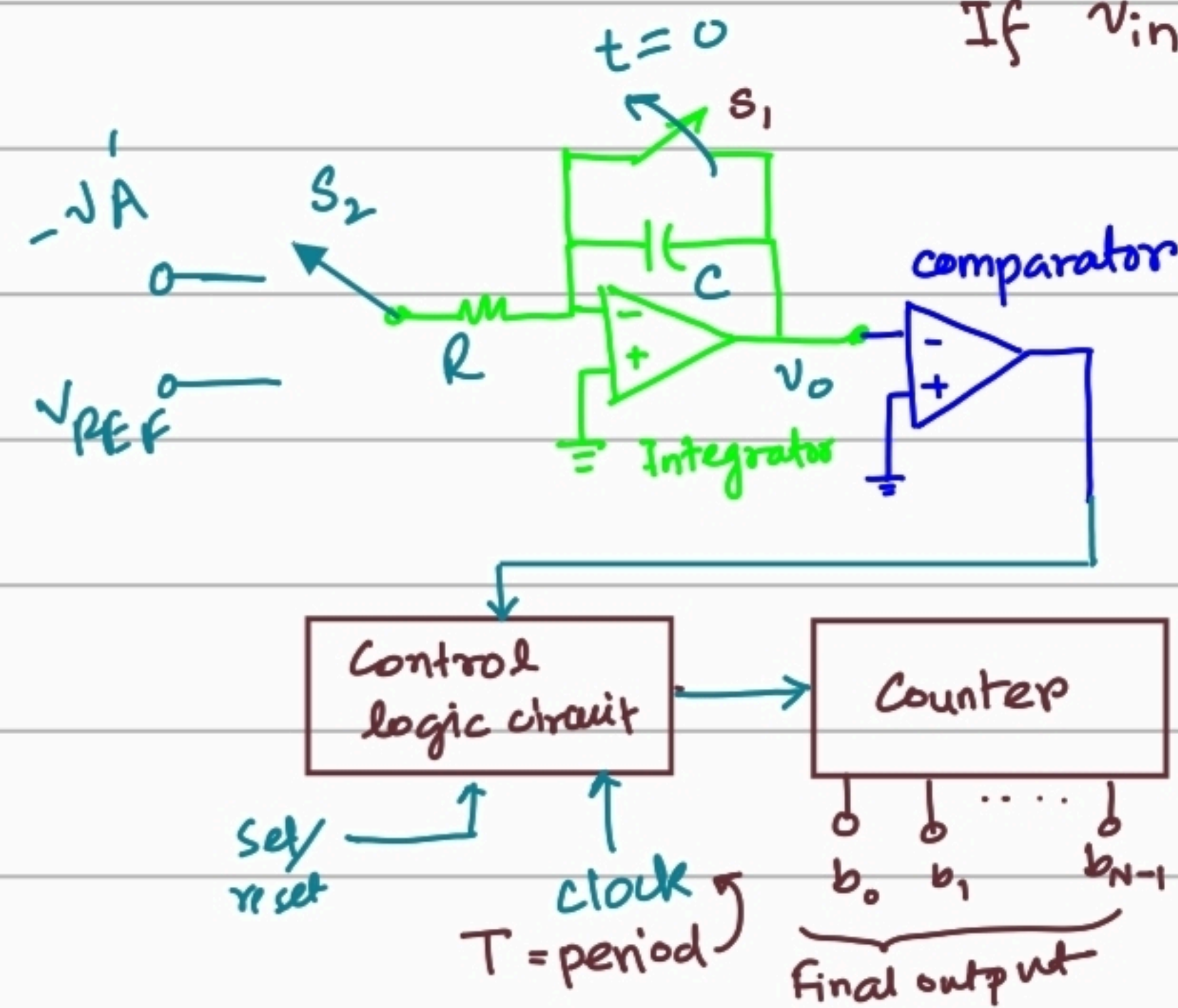
Then the output of the counter is the digital output we want.

Integrator:

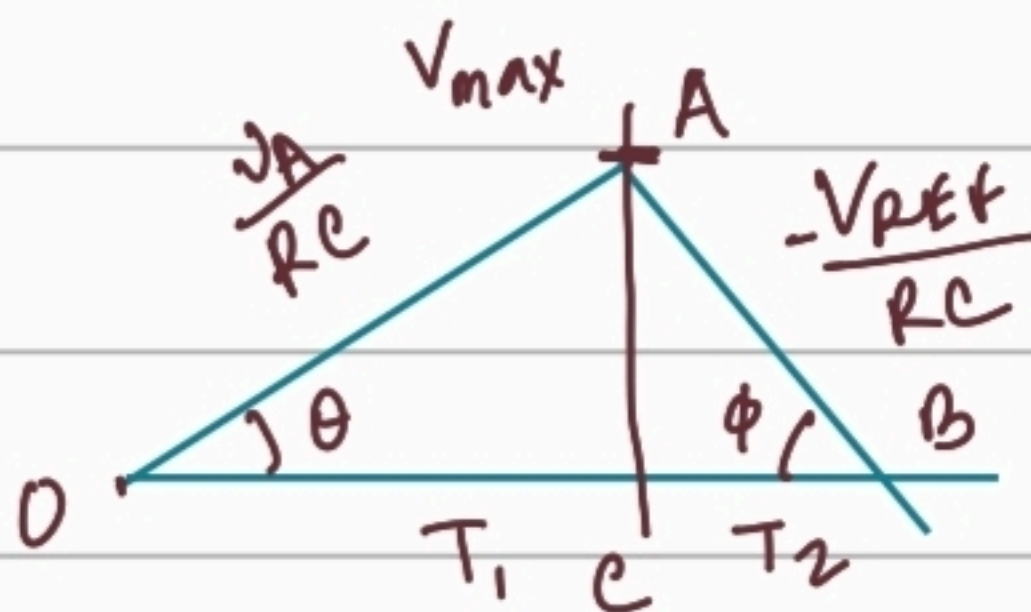


$\therefore \frac{dQ}{dt} = \text{current through the capacitor}$
 $= C \cdot \frac{dV_o}{dt} = i_c = \frac{0 - v_{in}}{R} \Rightarrow \boxed{\frac{dV_o}{dt} = -\frac{v_{in}}{RC}}$

If v_{in} is constant then $\frac{dV_o}{dt}$ is constant.



Say final output is binary representation of the number n .



$$\tan \theta = \frac{V_A}{RC} = \frac{AC}{OC} = \frac{V_{max}}{T_1}$$

$$\tan \phi = \frac{V_{REF}}{RC} = \frac{AC}{BC} = \frac{V_{max}}{T_2}$$

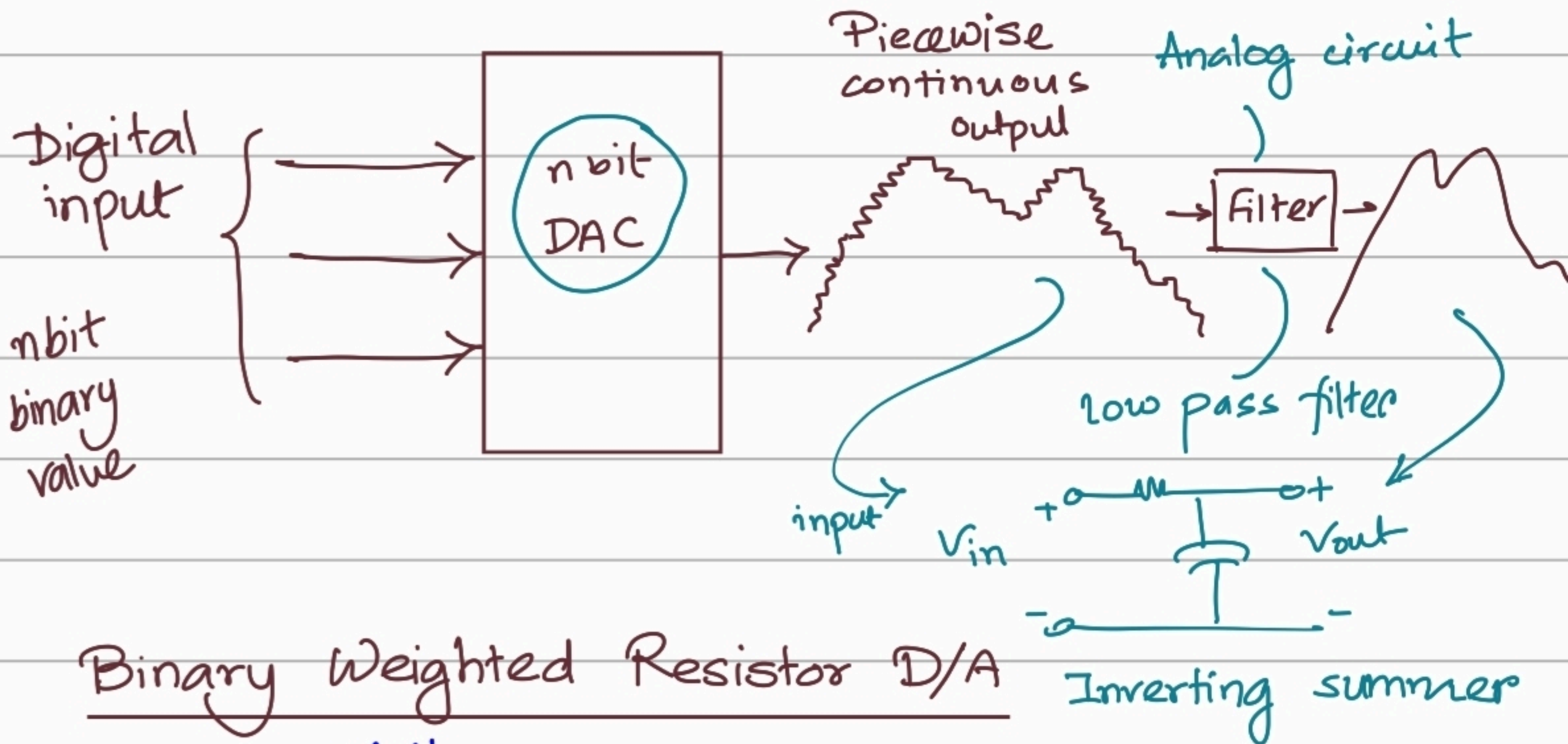
$$\frac{V_{max}/T_1}{V_{max}/T_2} = \frac{V_A}{V_{REF}} = \frac{T_2}{T_1} = \frac{nT}{2^N T}$$

$$\boxed{n = V_A \frac{2^N}{V_{REF}}}$$

$$\Leftarrow \boxed{V_A = \frac{n}{2^N} V_{REF}}$$

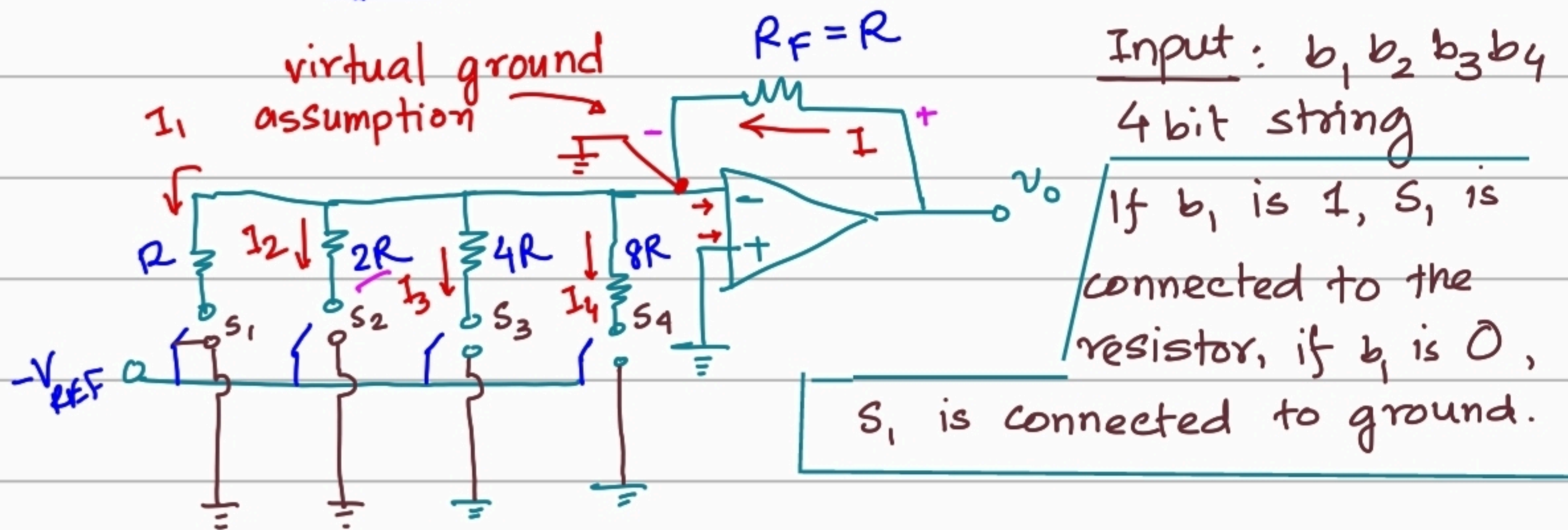
lecture 17: D/A and A/D converter

Digital to Analog Conversion



Binary Weighted Resistor D/A

4-bit



Same rules applies to S_2, S_3 and S_4 .

$$I = I_1 + I_2 + I_3 + I_4 = \frac{V_0 - 0}{R_F} = \frac{V_0}{R_F}$$

	I_1	I_2	I_3	I_4
switch is connected to resistor	$\frac{0 - (-V_{REF})}{R} = \frac{V_{REF}}{R}$	$\frac{V_{REF}}{2R}$	$\frac{V_{REF}}{4R}$	$\frac{V_{REF}}{8R}$
switch is connected to ground	0 because the path is open	0	0	0

$$\frac{V_0}{R_F} = b_1 \frac{V_{REF}}{R} + b_2 \frac{V_{REF}}{2R} + b_3 \frac{V_{REF}}{4R} + b_4 \frac{V_{REF}}{8R}$$

$$\Rightarrow V_0 = V_{REF} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right)$$

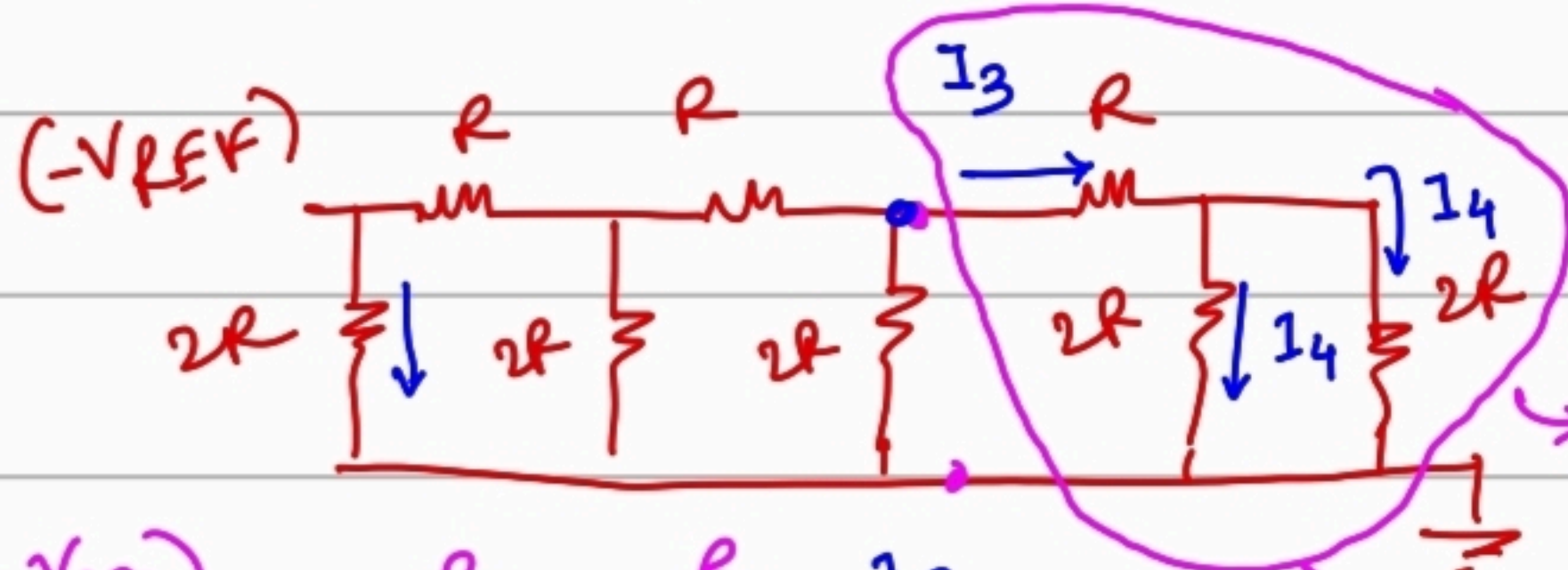
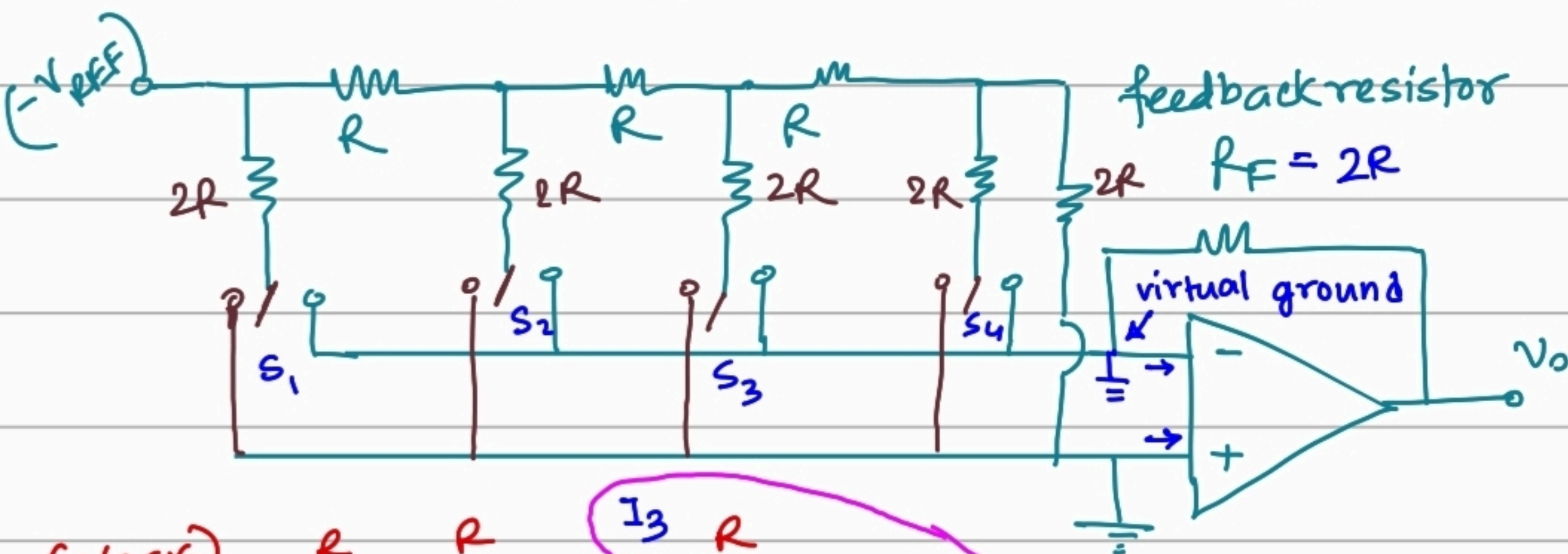
Design 6 bit BWR D/A where $V_{max} = 10V$.

$$V_o = V_{max} = V_{REF} (1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32})$$

$$V_{REF} \approx 5V$$

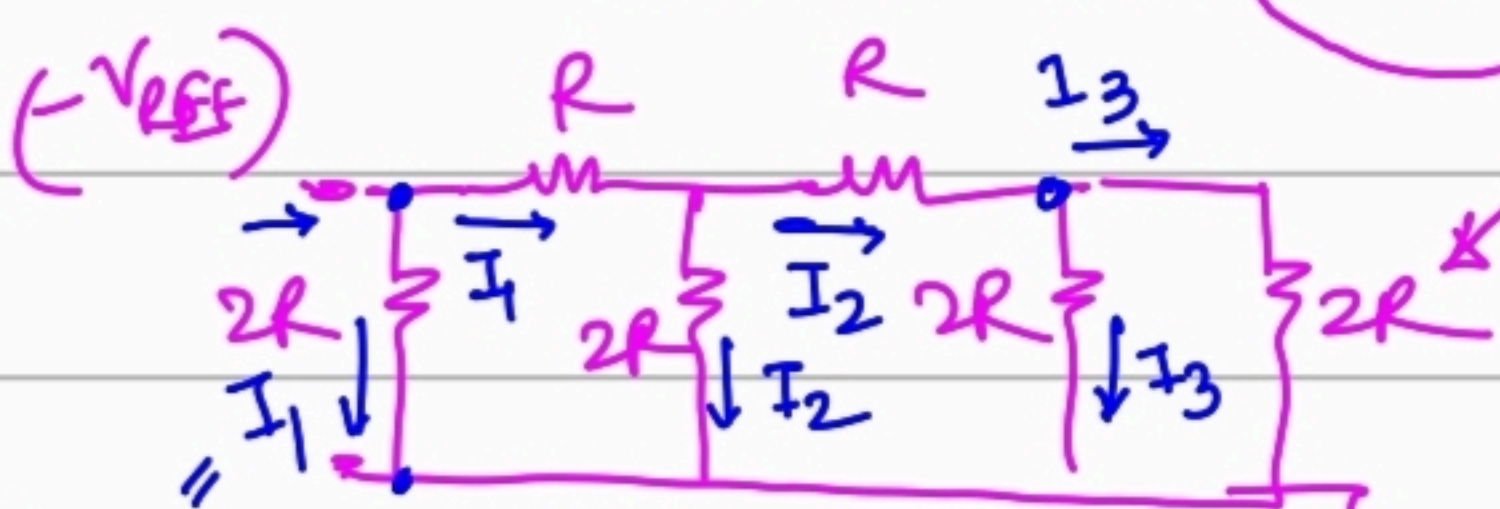
Dis: Output is dependent on value of the resistors and it is hard to find resistor in this much variations.

R-2R ladder: 4 bit R-2R ladder



$$2I_4 = I_3$$

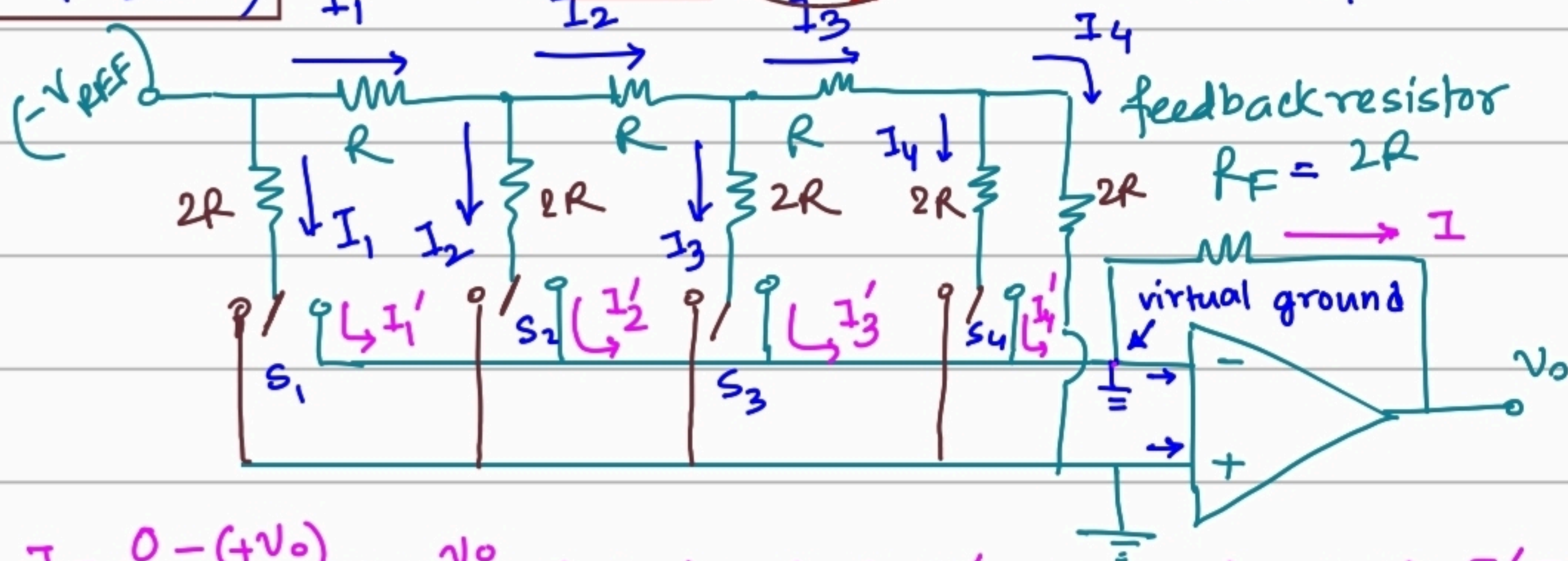
equivalent resistance
 $(2R || 2R) + R = 2R$



$$2I_3 = I_2$$

Similar if we continue this arguments, then we will get,
 $2I_2 = I_1$

$$I_1 = \left(\frac{-V_{REF} - 0}{2R} \right), \quad I_2 = \frac{I_1}{2}, \quad I_3 = \frac{I_2}{2}, \quad I_4 = \frac{I_3}{2}$$



$$I = \frac{0 - (+V_o)}{R_F} = -\frac{V_o}{R_F}, \text{ here } b_1 = 1 \Rightarrow I'_1 = I_1 \text{ or } b_1 = 0 \Rightarrow I'_1 = 0.$$

$$\text{So } I'_1 = b_1 I_1, \text{ similarly } I'_2 = b_2 I_2, I'_3 = b_3 I_3, I'_4 = b_4 I_4$$

$$\text{Thus } I = I'_1 + I'_2 + I'_3 + I'_4 = b_1 I_1 + b_2 \frac{I_1}{2} + b_3 \frac{I_1}{4} + b_4 \frac{I_1}{8}$$

$$I = -\frac{V_o}{R_f} = I_1 \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right) = -\frac{V_{REF}}{2R} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right)$$

$$\Rightarrow V_o = V_{REF} \left(b_1 + \frac{b_2}{2} + \frac{b_3}{4} + \frac{b_4}{8} \right)$$

Performance specifications.

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| 1. Resolution (Δ) [Homework, chapter 16 Neamen] | |
| 2. Reference voltage | 5. Speed |
| 3. Settling time | 6. Error |
| 4. Linearity | |